

Deep Video Understanding with a Unified Multi-Modal Retrieval Framework

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ABSTRACT

In this paper, we propose a unified multi-modal retrieval framework to tackle two typical video understanding tasks, i.e., matching movie scenes and text descriptions, and scene sentiment classification. For the task of matching movie scenes and text descriptions, it is a natural multi-modal retrieval problem, while for the task of scene sentiment classification, the proposed framework aims at finding most related sentiment tag for each movie scene, which is also a multi-modal retrieval problem. By considering these two tasks as multi-modal retrieval problems, we propose a unified multi-modal retrieval framework, which can make full use of the models pre-trained on large scale multi-modal datasets, experiments show that it is critical for the tasks which have only hundreds of training examples. To further improve the performance on movie video understanding task, we also collect a large scale video-text dataset, which contains 427,603 movie-shot and text pairs. Experimental results validate the effectiveness of this dataset.

CCS CONCEPTS

• Computing methodologies → Scene understanding.

KEYWORDS

video understanding, multi-modal retrieval

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1 INTRODUCTION

With the increasing number of videos uploaded on the Internet, video understanding becomes an emergent and important task for automatic video processing. Recently, Deep Video Understanding (DVU) challenge [4] is proposed to analyse movie videos with deep learning methods. In this paper, we focus on Question 4 and Question 5 of scene-level queries, i.e., matching movie scenes and text descriptions, and scene sentiment classification. For the task of matching movie scenes and text descriptions, we need to find 1-to-1 relationship between movie scenes and natural language descriptions, for the task of scene sentiment classification, we need to classify the scene sentiment from a given scene.

One main challenge of the deep video understanding task is that it is expensive to obtain large scale training data, because annotating long movie videos is labor intensive. Although recently proposed HLVU dataset [4] releases hundreds of annotated movie shots, it is still not enough to train modern deep learning methods. Moreover, for the scene sentiment classification problem, several sentiment categories even don't appear in the training set, typical classification methods will fail for these cases.

To tackle the above challenges, we propose a unified multi-modal retrieval framework to solve video-text matching problem and scene classification problem simultaneously. Specifically, for the video-text matching problem, "Finding 1-to-1 relationship between scenes and natural language descriptions" is a natural video-text retrieval task. Meanwhile, for the sentiment classification problem, "Classifying scene sentiment from a given scene" can also be considered as a video-text retrieval task, i.e., find the most related sentiment tag for a given scene.

Tackling the video-text matching problem and scene sentiment classification problem by multi-modal retrieval has two main advantages: 1) Pre-trained multi-modal models that trained on large-scale dataset can be used for transfer learning, experiments show it is critical for better performance. 2) we can tackle the video-text matching problem and scene sentiment classification problem with one unified framework, which makes our system simple and clear.

Although there are a lot of pre-trained multi-modal models available [11], they are not trained on movie dataset. To further improve the performance of movie video understanding models, we collect a large scale video-text dataset, which consists of over 400,000 movie

shot and text pairs. Meanwhile, we develop an automatic data collection pipeline, which enables us to collect the large scale dataset with less manual labor.

The main contributions of this paper are summarized as follows.

- We propose a unified multi-modal retrieval framework to tackle video-text matching problem and sentiment classification problem simultaneously. Multi-modal information including video frames, audio and query text are taken into consideration for better performance.
- We construct a large-scale movie video-text dataset, which consists of 427,603 video-text pairs. The collected dataset is significantly larger than previous movie video-text datasets.
- Comprehensive experiments are conducted to validate the effectiveness of the proposed framework and the collected dataset.

2 RELATED WORKS

Movie video-text dataset. MovieQA [15] is a large question-answer dataset about movies, which aims to automatically understand movie content from video and text. It consists of 14,944 questions for 408 movies, each question has 5 candidate answers while only one is correct. MPII-MD [12] contains 94 Hollywood movies that are segmented into 68,337 clips with an average duration of 3.9 seconds, each clip has a corresponding sentence. M-VAD [16] is generated by Descriptive Video Service (DVS), which consists of 48,986 video clips from 92 movies. A large scale dataset and challenge (LSMDC) [13] includes over 128,000 video-sentence pairs.

Vision-language model. Vision-language models have gained a lot of attention recently [2, 3, 5–8, 17, 18], they aim at establishing relationship between visual data (e.g., image and video) and text data. State-of-the-art methods usually use large scale dataset for training, e.g., CLIP [18] uses 400 millions of image-text pairs for pre-training, and ALIGN [5] uses 1.8 billions of image-text pairs. However, the video-text datasets are relatively smaller than image-text datasets because collecting video-text data is more expensive. Previous method like CLIP4CLIP [9] propose to use pre-trained CLIP model to initialize their encoders, then fine-tune their framework on video-text datasets.

3 DATASET COLLECTION

We collect a large-scale video-text dataset with 427,603 video-text pairs. For each video-text pair, the text sentence describes the content of corresponding video. The dataset is constructed with the help of the Audiovault project¹. The Audiovault project provides audio descriptions for about 9000 movies, the audio descriptions can be easily convert to text descriptions through Automatic Speech Recognition (ASR), which means that we can obtain a lot of movie level video-text data with only a little manual labor. However, the video-text correspondence in movie level data is too coarse, it is better for us to establish more accurate correspondence between video and text, e.g., obtain corresponding video scene for each text sentence. To obtain accurate sentence level video-text pairs, we need to solve two problems:

- Movies may have different trimmed versions, so the movie videos we collected may be different with those used to create audio descriptions. We need to filter these mismatched audio description and movie video pairs.
- The audio description keeps conversation of the original movie, which should be removed because we only need the description content.

For the first problem, following previous work [14], we sample a ten-minute video clip starting from the t_0 -th second of the video, then extract its audio signal, we denote this audio signal as $f(t)$, for the corresponding audio description, we denote its audio signal as $g(t)$. If the audio description and the movie video are matched, then we will have:

$$t_0 = \operatorname{argmax}_{t_\Delta} f(t) \star g(t + t_\Delta). \quad (1)$$

where $\star$ is the cross-correlation operator. In order to make the verification more reliable, we sample 10 ten-minutes video clips and conduct the verification, respectively. We will keep the movie video and corresponding audio description only if all the 10 verification are passed.

For the second problem, we first conduct ASR for both the audio description and the movie video, resulting in M text sentences for the audio description and N text sentences for the movie video. Start timestamps and end timestamps for text sentences of audio description are denoted as $(s_i, e_i)_{i=1}^M$, start timestamps and end timestamps for text sentences of movie video are defined as $(s'_j, e'_j)_{j=1}^N$, respectively. For each (s_i, e_i) , we remove it if the IoU metric between it and any $(s'_j, e'_j)_{j=1}^N$ is larger than 0.5, the IoU metric is calculated by following

$$IoU = \frac{\min(e_i, e'_j) - \max(s_i, s'_j)}{\max(e_i, e'_j) - \min(s_i, s'_j)} \quad (2)$$

After all the processes described above, we obtain 427,603 video clips from 1624 movies. By comparing the names of these 1624 movies and the 6 test movies of DVU Challenge 2022, we confirm that the test movies are not included in the collected dataset.

4 METHOD

In this section, we introduce our multi-modal framework, which aims at tackling video-text matching task and sentiment classification task in a unified framework.

4.1 Framework

As shown in Figure 1, our framework takes three types of information as inputs: the video frames, the audio signal, and a text query. Both the frames and the audio signal are extracted from the input video. The content of the input text query depends on the task.

For the video-text matching task, it is a sentence which describes the content of corresponding video. Given the input video and the text query, the framework first encode them into feature space, then employs the matching module to predict the similarity between input video and text.

For the sentiment classification task, we consider it as a video-to-text retrieval task, i.e., given the input video and a set of candidate

¹<https://audiovault.net/>

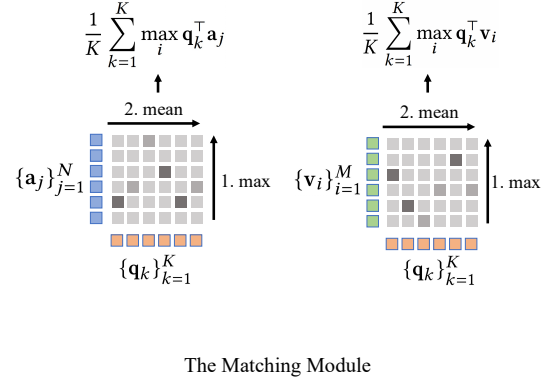
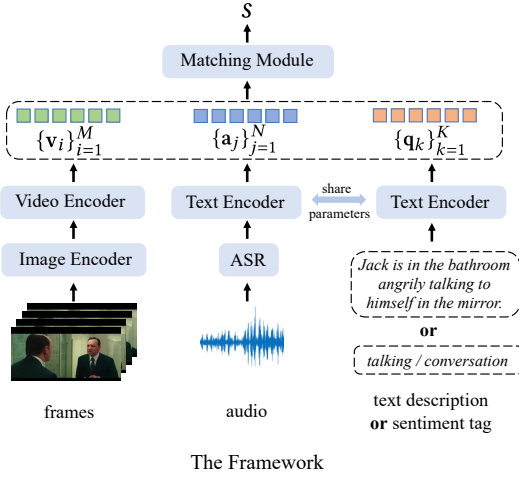


Figure 1: Left: The multi-modal retrieval framework introduced in Section 4.1. Right: The implementation details of the matching module described in Section 4.5

sentiment tags, we need to predict which sentiment tag best describes the video. Considering the sentiment classification task as a video-to-text retrieval task brings two practical advantages:

- The unified framework can be initialized by recently released large scale pre-trained multi-modal models, e.g., CLIP, which benefits us a lot because for the two tasks we only have hundreds of training data.
- We can build a unified multi-modal retrieval framework to tackle video-text matching task and sentiment classification task simultaneously, which makes our system simple and clear.

Considering the success of the prompt techniques [20], we also apply prompt before we feed the sentiment tag into the framework, e.g., for a given sentiment tag *talking/conversation*, we convert it into a sentence *This is a frame of talking/conversation*.

For the input frames, audio and text, the framework employs a visual branch, an audio branch and a text branch to encode them into feature embeddings before matching the videos and the queries.

4.2 The Visual Branch

The visual branch extracts feature representation for the video frames, it consists of an image encoder and a video encoder. The image encoder encodes input frames into embedding vectors $\{f_i\}_{i=1}^M$, where f_i indicates the embedding vector of the i -th frame. However, encoding each frame individually will ignore the relationship between the frames, so we introduce a video encoder to propagate the information between frames following [9]. The video encoder is a neural network consists of several transformer blocks, it takes $\{f_i\}_{i=1}^M$ as input, conducts temporal feature propagation and produces $\{v_i\}_{i=1}^M$ as output.

4.3 The Audio Branch

For the input audio signal, the most important information is the conversation between roles in the movie. We extract the conversation information by Automatic Speech Recognition (ASR), and save it into text sentences. After that, we tokenize the text sentences,

and feed them into a text encoder to get the audio representation $\{a_j\}_{j=1}^N$, where a_j is the embedding of the j -th token.

4.4 The Query Branch

As we described in Section 4.1, for both the video-text matching task and the sentiment classification task, we formulate the input query as text sentences. This text sentence will be tokenized and fed into a text encoder, the text encoder produces token-level embeddings $\{q_k\}_{k=1}^K$. Note that the text encoder in the query branch shares the same architecture and parameters with that one in the audio branch.

4.5 The Matching Module

After we get the representation of input frames, audio and query. We can compute the similarity between input video and query with the matching module. The matching module first L2-normalizes $\{v_i\}_{i=1}^M$, $\{a_j\}_{j=1}^N$ and $\{q_k\}_{k=1}^K$, then compute the final matching score following FILIP [19]:

$$s = \frac{1}{K} \sum_{k=1}^K \max_i q_k^T v_i + \frac{1}{K} \sum_{k=1}^K \max_i q_k^T a_j. \quad (3)$$

where s is the similarity score between input video and query. During the training stage, the framework can be trained end-to-end with the following loss function:

$$\mathcal{L} = -\log\left(\frac{\exp(s_p/\tau)}{\sum_i \exp(s_i/\tau)}\right). \quad (4)$$

where s_p is the similarity between a matched video and query, and s_i is the similarity between the query and all videos in current batch, τ is the temperature factor, which is initialized as 0.07 and trained end-to-end.

5 EXPERIMENTS

5.1 Implementation Details

We implement our framework by PyTorch [10]. Both the image encoder and the text encoder are 12-layers transformers [17], which

Table 1: Experiments on the video-text matching task. *zs* indicates zero-shot evaluation, i.e., we directly evaluate the model trained on image-text data on DVU dataset. *pt* indicates pre-training the model on our collected dataset, and *ft* represents fine-tuning the model on DVU dataset.

Movies	Video-text matching (%)			
	CLIP4CLIP <i>zs</i>	CLIP4CLIP <i>pt</i>	Ours <i>pt</i>	Ours <i>ft</i>
Spiritual Contact	65.8	68.4	84.2	92.1
Honey	26.1	30.4	47.8	69.6
Nuclear Family	62.5	68.8	100.0	87.5
Sophie	26.7	15.6	24.4	37.8
Superhero	60.0	100.0	80.0	80.0
Huckleberry Finn	16.9	32.2	45.8	57.6
Shooters	50.0	65.0	70.0	90.0
The Big Something	42.4	53.0	65.2	74.2
Time Expired	42.7	40.0	73.3	81.3
Valkaama	66.7	73.3	88.9	80.0
Average	46.0	54.7	68.0	75.0
Test Movies	-	-	-	100.0

Table 2: Experiments on the sentiment classification task.

Movies	Sentiment(%)	
	Baseline	Ours
Spiritual Contact	50.0	86.8
Honey	87.0	40.9
Nuclear Family	43.8	73.3
Sophie	42.2	68.8
Superhero	10.0	60.0
Huckleberry Finn	56.0	86.4
Shooters	75.0	30.0
The Big Something	51.5	38.1
Time Expired	64.0	78.1
Valkaama	68.9	73.3
Average	54.9	63.6
Test Movies	-	61.0

are initialized by the pre-trained CLIP model [18] and trained with a small initial learning rate equal to $1e-7$. The video encoder is a 4-layers transformer, which is random initialized and trained with an initial learning rate equal to $1e-4$. The learning rate of the entire framework is decayed following a cosine schedule. For all experiments in this paper, we evaluate the model on the same 10 movies as previous works [1], the evaluation metric is calculated as:

$$Score = \frac{\#CorrectAnswers}{\#TotalQuestions}. \quad (5)$$

as described in previous work [4].

5.2 Pre-Training

We first take CLIP4CLIP [9] as baseline model for the video-text matching task. We use the parameter-free type of CLIP4CLIP in

order to conduct zero-shot inference on the challenge dataset, the model is denoted by CLIP4CLIP *zs*, results are shown in the first column of Table 1.

We further pre-train the CLIP4CLIP baseline on the collected dataset described in Section 3, the model is denoted by CLIP4CLIP *pt*, results are shown in the second column in Table 1. We can see that pre-training on the collected dataset improves the CLIP4CLIP baseline by 8.7%, which validates the effectiveness of the collected dataset.

After that, we pre-train our framework introduced in Section 4.1 on the collected dataset, the model is denoted by Our *pt*, results are shown in the 3rd column of Table 1. By introducing audio information and more advanced matching module described in Section 4.5, our framework outperforms CLIP4CLIP baseline by 13.3%.

5.3 Fine-tuning on DVU dataset

We further fine-tune our framework on DVU dataset. Since DVU dataset has no official validation set, we conduct 7-folds cross-validation experiments on the total 14 movies, i.e., for each fold we use 12 movies for training and the other 2 movies for evaluation. Although we will have evaluation results for all 14 movies, we only report results on the same 10 movies as Section 5.2 for comparison.

Video-text matching task The results is shown in the 4-th column of Table 1, by fine-tuning our framework on DVU dataset, we further obtain 7.0% improvements.

Sentiment classification task For the sentiment classification task, we first develop a classification baseline, it contains two components: a video feature extractor and a linear classifier. The video feature extractor is initialized by the visual branch of the pre-trained model described in Section 5.2, features of different frames are averaged. We then fine-tune and evaluate the classification baseline, the results is shown in the first column of Table 2. Meanwhile, we fine-tune our framework on the DVU dataset for sentiment retrieval, the results are shown in the second column of Table 2. Our proposed framework outperforms the classification baseline by 8.7%.

More discussion After the cross-validation experiments described above, we further train our framework on all training data of DVU, the final results on test movies are shown in Table 1 and Table 2. The proposed model obtains 100% matching accuracy on the test movies, much better than our validation results, it may be because that the test task (matching 10 video scenes and 10 description texts for each movie) is more easier than our validation task (matching at most 70+ video scenes and description texts for each movie).

6 CONCLUSION

In this paper, we consider the video-text matching task and the sentiment classification task as multi-modal retrieval problems, and proposed a unified framework to tackle both tasks simultaneously. This framework can make full use of the vision-language models pre-trained on large scale datasets, e.g., CLIP. Experiments in Section 5 show its effectiveness. Besides that, we also collect a large scale video-text dataset, which contains 427,603 video-text pairs from 1624 movie videos. The collected dataset significantly improves the performance of our proposed framework.

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